

From WCM to SpaceTime: Fixed-Query Visual History Encoding for Robot Value Prediction

WCM Project

October 2026

Abstract

Visual histories can resolve state ambiguity in robot value prediction, yet their length increases the input sequence seen by the critic. We develop SPACETIME, an extension of the World Critic Model (WCM) that encodes eight sampled frames from two cameras through a shared vision backbone and temporal interaction layers, then reduces the history to 64 learned queries. Holding the number of frames, queries, model blocks, and training schedule fixed, we compare histories spanning 60 and 120 frame offsets. Across three seeds on a 125-episode out-of-distribution (OOD) set, the 60-offset system reduces paired value-prediction error relative to the original WCM. Extending the span to 120 offsets further lowers paired OOD mean squared error for all three seeds, with episode-bootstrap 95% intervals excluding zero. An episode-level success/failure diagnostic on an 18-episode checkpoint-validation split additionally shows 18/18 correct predictions at frame 2 for each SPACETIME seed. These results support fixed-query encoding of a longer sampled history in the evaluated robot task; they do not imply fixed inference cost or a new temporal-attention operation.

1 Introduction

The value of a robot state depends on progress that may be ambiguous in a single image. A short sequence can expose motion and interaction context, but carrying more frames through a critic also increases the visual input. The original WCM uses local history for action-free value prediction, whereas its visual interface does not explicitly compress a longer sampled history into a fixed-size set of tokens. This study asks whether a fixed latent interface can make a longer visual history useful for value prediction.

SPACETIME processes eight observations at selected historical offsets. A shared ViT encodes each camera frame, temporal layers mix information across observed times, and Perceiver-style cross-attention reduces the resulting sequence to 64 learned queries [1, 2]. The downstream WCM consumes this fixed-size representation. We test the route from the original WCM to an eight-frame history at maximum offset 60, then change only the sampling offsets to cover a maximum offset of 120. The main experiment therefore isolates temporal span within the SPACETIME system rather than proposing a new attention primitive.

The evidence follows three steps. First, we compare original WCM and SPACETIME on paired OOD endpoints. Second, we compare the two SPACETIME spans under the same model and evaluation protocol. Third, we inspect the model’s trained failure-risk output on an episode-disjoint validation split and place it beside a value-based diagnostic from the original WCM. The latter uses different prediction heads and is interpreted as a system-level comparison.

2 Method

2.1 Original WCM and prediction targets

The original WCM predicts an action-free state value $\hat{V}_t = f_V(\mathcal{H}_t, p_t)$ from its short local history $\mathcal{H}_t$ and proprioceptive state p_t . Action is consumed by the separate $Q(s, a)$ and action-conditioned dynamics paths after the value path. The SPACETIME systems retain value, Q , and dynamics heads and also train a failure-risk head. Its logit r_t uses the target $y_t^{\text{fail}} = 1 - y_t^{\text{success}}$, so $\sigma(r_t)$ is a failure score. The original WCM checkpoint has no risk head; its failure diagnostic below is constructed from $-\hat{V}_t$.

2.2 Fixed-query history encoder

For a maximum sampled offset k , let

$$\mathcal{X}_t^{(k)} = (x_{t-\delta_1}, \dots, x_{t-\delta_8}), \quad \max_i \delta_i = k. \quad (1)$$

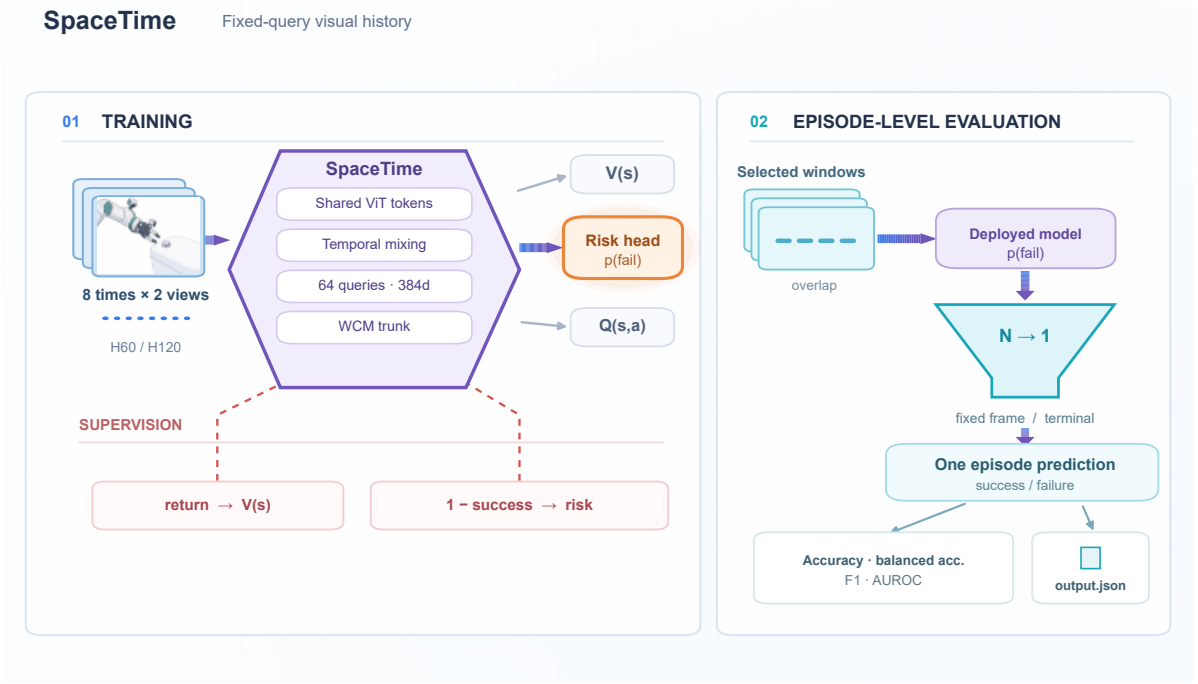


Figure 1: SpaceTime training and episode-level evaluation. Eight sampled time points from two cameras enter the shared ViT, temporal layers and 64-query bottleneck. Value is supervised by return; the failure-risk head uses $1 - \text{success}$. At evaluation, overlapping windows feed the deployed model and one fixed-frame or terminal prediction is selected per episode before computing classification metrics and writing the JSON report. The input-card illustration is AI-generated concept art, not an experimental video frame. The risk head belongs to the SPACETIME checkpoints, not to the original WCM checkpoint.

We denote this eight-frame history by H_k . The two offset lists are

$$H_{60} : [0, 9, 17, 26, 34, 43, 51, 60], \quad (2)$$

$$H_{120} : [0, 17, 34, 51, 69, 86, 103, 120]. \quad (3)$$

Offsets are frame indices, not elapsed seconds or numbers of input frames. An offset preceding the episode start is clamped to the first frame, so the earliest predictions repeat that frame in several historical slots.

Each of the two camera views is passed through a shared ViT ϕ . With 224-pixel inputs, each view yields one CLS token and 196 patch tokens. Eight frames from two cameras therefore produce $8 \times 2 \times 197 = 3152$ visual tokens before reduction. The SPACETIME representation is

$$h_t^{(k)} = \Pi_{64} \left(\Psi_{\theta} \left(\phi(\mathcal{X}_t^{(k)}) \right) \right), \quad \hat{V}_t^{(k)} = g_V(h_t^{(k)}, p_t), \quad (4)$$

where Ψ_{θ} contains two factorized temporal layers and Π_{64} contains two Perceiver-style reduction layers. The 64 output queries each have width 384. The vision backbone is trainable and the CLIP text features are frozen [3]. Figure 1 shows the input and fixed-query interface. Here “fixed” describes the 64-query latent bottleneck: input pixels, FLOPs, and wall-clock time are reported separately.

3 Experimental protocol

3.1 Models and value evaluation

The experimental route is original WCM $\rightarrow$ T8- H_{60} SPACETIME $\rightarrow$ T8- H_{120} SPACETIME. The two SPACETIME variants have the same architecture, 64-query bottleneck, same-seed Image-B4 initialization, and two-stage schedule: ten joint epochs followed by three full fine-tuning epochs. Their only changed model input is the offset list. The original WCM checkpoint uses a different training recipe and supports a system-level, rather than component-isolated, comparison.

The 0722 OOD set (125 episodes) is the primary value-prediction evaluation; the 0814 5-cut set supports near-distribution analysis. Paired comparisons use identical episode/frame endpoints and 20,000 episode-level

bootstrap resamples per seed. Mean squared error (MSE) is primary, while Pearson correlation, centered MSE, and mean bias diagnose ordering and calibration. The original WCM versus H_{60} comparison uses 21,832 common OOD endpoints per seed. The H_{60} versus H_{120} comparison uses a separate manifest of 22,082 endpoints per seed. Absolute metrics are only compared within their respective manifests.

3.2 Episode success/failure diagnostic

The expanded-data checkpoint-validation split contains 18 episodes: 13 successes and five failures. We select one prediction per episode at frame index 2, the earliest common frame for the original WCM and SPACETIME checkpoints. For SPACETIME, a risk score $\sigma(r_t) \geq 0.5$ predicts failure. For original WCM, the score is $-\hat{V}_t$; its threshold is selected on training episodes to maximize balanced accuracy and then frozen on validation episodes. We report correct episodes, failure recall, balanced accuracy, macro-F1, AUROC, and the underlying confusion counts. The same 18 episodes are reused across model seeds and were part of checkpoint selection, so the seed results are repeated model evaluations of one validation cohort.

Risk trajectories additionally sample one prediction per episode at fractions 0, 0.25, 0.5, 0.75, 1 of the available window sequence. These fractions are computed from completed episode lengths for analysis; the fixed frame-2 comparison is the operational early-time readout. All five validation failures come from the 0703 failure collection, while 11 of 13 successes come from an 0814 success tail. Collection-related appearance is therefore a possible contributor to risk separation on this split.

4 Results

4.1 Explicit visual history improves OOD value prediction

On the 21,832-endpoint common OOD manifest, T8- H_{60} reduced MSE relative to original WCM in each of three seeds (Table 1; Fig. 2). The paired MSE changes were -0.11833 (95% CI $[-0.12618, -0.11060]$), -0.08568 ($[-0.09314, -0.07849]$), and -0.09157 ($[-0.09922, -0.08403]$) for seeds 3072, 42, and 1337. Pearson correlation increased for all three seeds. This establishes a repeatable system-level gain from the longer-history encoder under the evaluated OOD protocol.

Table 1: Original WCM and T8- H_{60} on common OOD endpoints. Each seed uses the same 125 episodes and 21,832 endpoint predictions. MSE reduction is $(\text{MSE}_{\text{WCM}} - \text{MSE}_{H_{60}})/\text{MSE}_{\text{WCM}} \times 100\%$.

Seed	Original WCM		T8- H_{60}		MSE reduction
	MSE	Pearson r	MSE	Pearson r	
3072	0.15167	0.140	0.03334	0.838	78.02%
42	0.11036	0.226	0.02467	0.851	77.64%
1337	0.10926	0.258	0.01769	0.835	83.81%

4.2 A longer span reduces error at fixed frame and query counts

The controlled $H_{60} \rightarrow H_{120}$ comparison uses eight input frames and 64 output queries for both variants. On 22,082 paired OOD endpoints, extending the maximum offset to 120 reduced MSE in all three seeds (Table 2; Fig. 3). The paired changes ranged from -0.00399 to -0.00688 , and every 95% CI excluded zero. Pearson gains were clear for seeds 3072 and 1337; seed 42 had a positive point estimate whose interval crossed zero. The larger span therefore supports the paper’s fixed-query history claim through consistent MSE improvements within the tested offset range.

Table 2: T8- H_{120} minus T8- H_{60} on paired OOD endpoints. Each seed has 125 episodes and 22,082 common endpoints. Intervals are 95% episode-level paired bootstrap intervals from 20,000 resamples. MSE reduction is $(\text{MSE}_{H_{60}} - \text{MSE}_{H_{120}})/\text{MSE}_{H_{60}} \times 100\%$ on these paired endpoints.

Seed	ΔMSE [95% CI]	MSE reduction	$\Delta\text{Pearson } r$ [95% CI]
3072	-0.00688 $[-0.00887, -0.00504]$	20.48%	$+0.02968$ $[+0.01838, +0.04340]$
42	-0.00399 $[-0.00549, -0.00257]$	16.01%	$+0.00449$ $[-0.00584, +0.01639]$
1337	-0.00420 $[-0.00596, -0.00261]$	23.68%	$+0.02695$ $[+0.01234, +0.04407]$

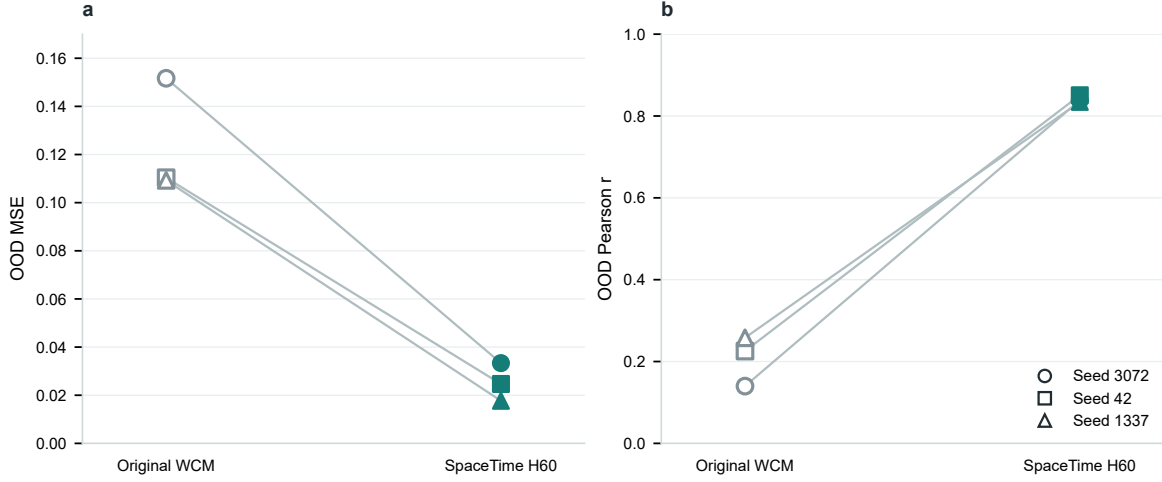


Figure 2: OOD value prediction across three model seeds. Paired lines connect the original WCM and T8- H_{60} checkpoint from the same seed. (a) Mean squared error. (b) Pearson correlation. All points use the same 125 episodes and 21,832 common endpoints per seed. Markers identify seeds 3072 (circle), 42 (square), and 1337 (triangle).

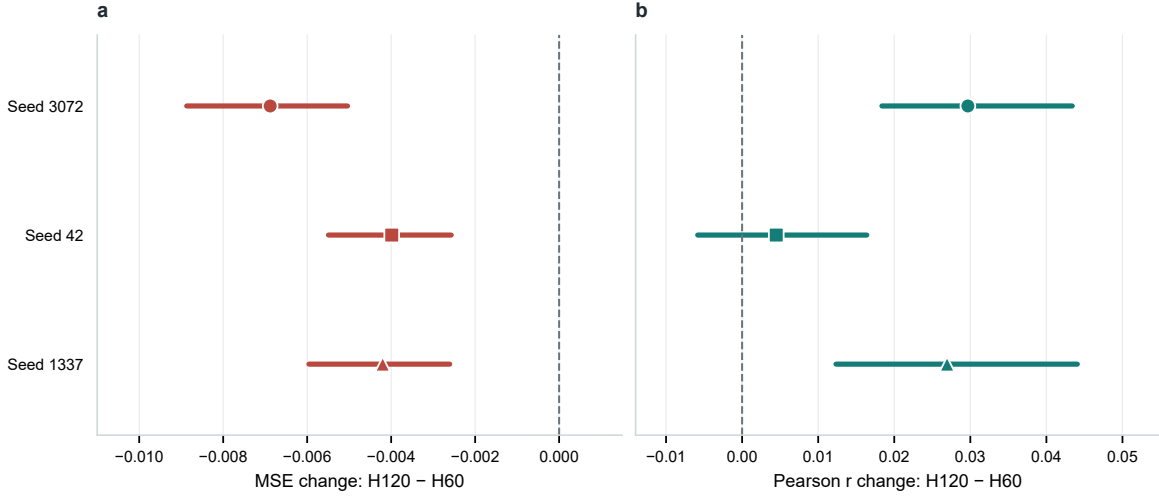


Figure 3: Paired effect of extending the temporal span. Seed-level point estimates and 95% episode-bootstrap intervals for T8- H_{120} minus T8- H_{60} : (a) MSE and (b) Pearson correlation. The dashed line denotes zero change. Both variants are evaluated on identical endpoints within each seed; these 22,082 endpoints differ from those in Fig. 2.

4.3 Failure scores on checkpoint validation episodes

At the shared frame-2 readout, both SPACETIME spans classified all 18 validation episodes correctly for each of three seeds. The original WCM value diagnostic classified 11, 13, and 13 episodes correctly for seeds 3072, 42, and 1337, respectively (Fig. 4). Its failure recall was 0/5, 0/5, and 1/5, compared with 5/5 for each SPACETIME checkpoint. The contrast describes the deployed systems and their respective score heads; it does not isolate the temporal encoder from the trained risk head.

Figures 5 and 6 trace the failure score over five episode-progress points and show its distribution across individual validation episodes. Figure 7 summarizes threshold-free discrimination at the fixed frame-2 readout: the mean original-WCM AUROC across seeds is 0.677 (95% paired-episode bootstrap interval [0.446, 0.872]), whereas the H_{120} risk score separates the observed validation labels completely (AUROC and average precision both 1.0). Figure 8 retains the individual scores at three progress points. Progress percentages index completed episode duration for retrospective visualization, whereas frame 2 is the fixed-time comparison.

4.4 Qualitative video observations

The supplied video artifacts show predicted value curves over an OOD trajectory and a separate failure episode. Figure 9 displays three frames from each example; the two examples were selected from available

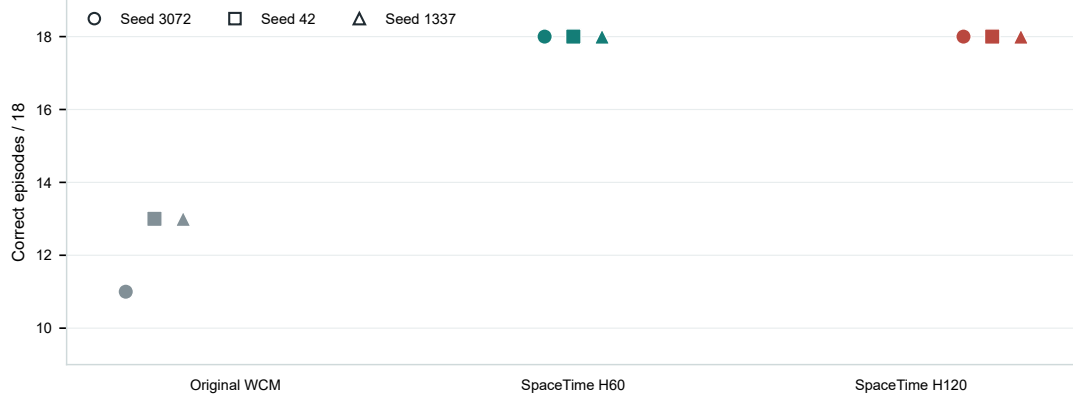


Figure 4: Episode-level success/failure diagnostic at frame 2. Correct validation episodes out of 18 (13 true successes, five true failures), shown separately for model seeds 3072, 42, and 1337. Original WCM uses $-V(s)$ with a seed-specific threshold calibrated on training episodes. SPACETIME H_{60} and H_{120} use their trained failure-risk heads with a score threshold of 0.5. The cohort is shared across seeds and contains collection-related outcome imbalance.

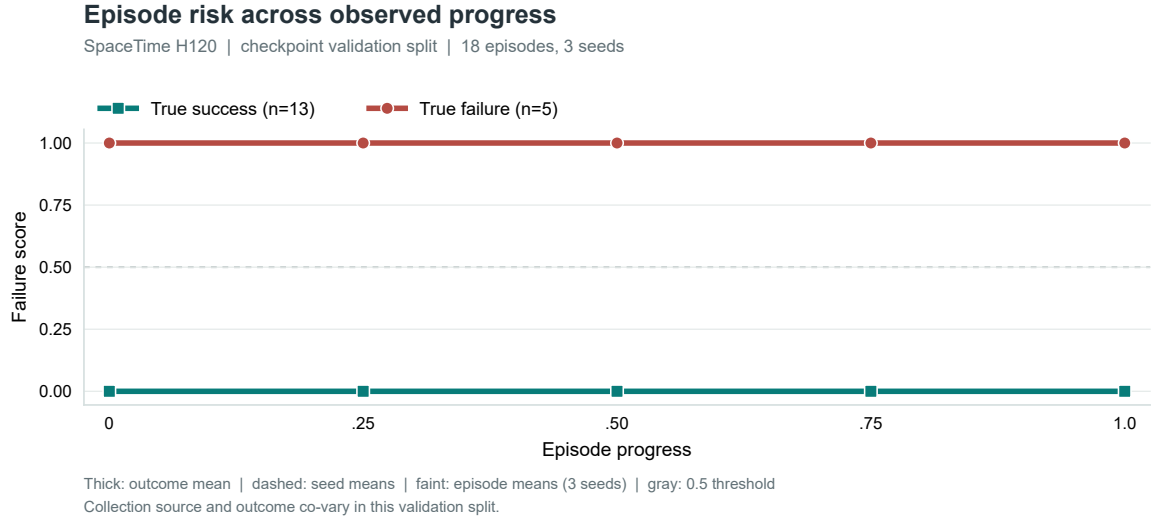


Figure 5: Failure-risk trajectories across episode progress. True-success and true-failure episodes are summarized at fractions 0, 0.25, 0.5, 0.75, and 1 of the available episode windows. The score is $\sigma(r_t)$ from T8- H_{120} . Curves summarize 13 successes and five failures from the same validation cohort; seeds are repeated model fits, not independent episodes. Progress fractions use the completed episode length.

rendered artifacts and are descriptive rather than a selection over the entire evaluation set. The full animations are available as Supplementary Videos S1 and S2.

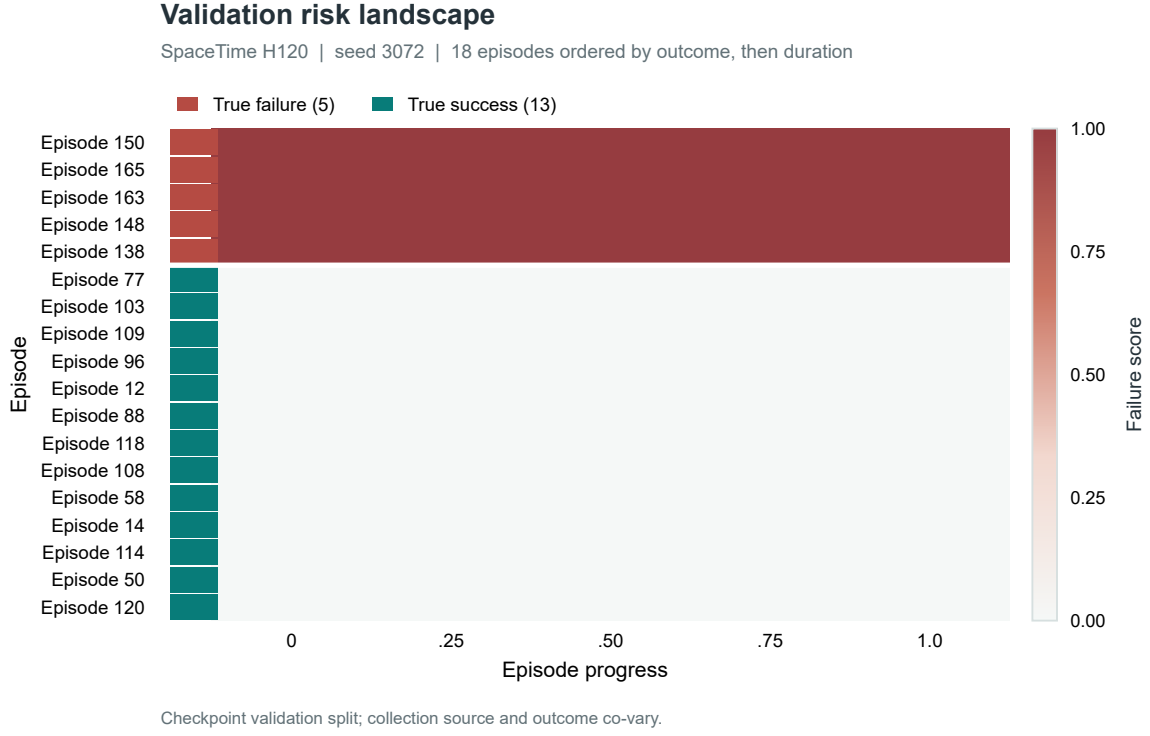


Figure 6: Episode-level failure-risk landscape. Rows are the 18 validation episodes for a representative T8- H_{120} seed; columns are the five episode-progress fractions. Color encodes $\sigma(r_t)$, and the outcome strip at left indicates true success or failure. Rows are grouped by true outcome and ordered by episode length within each group.

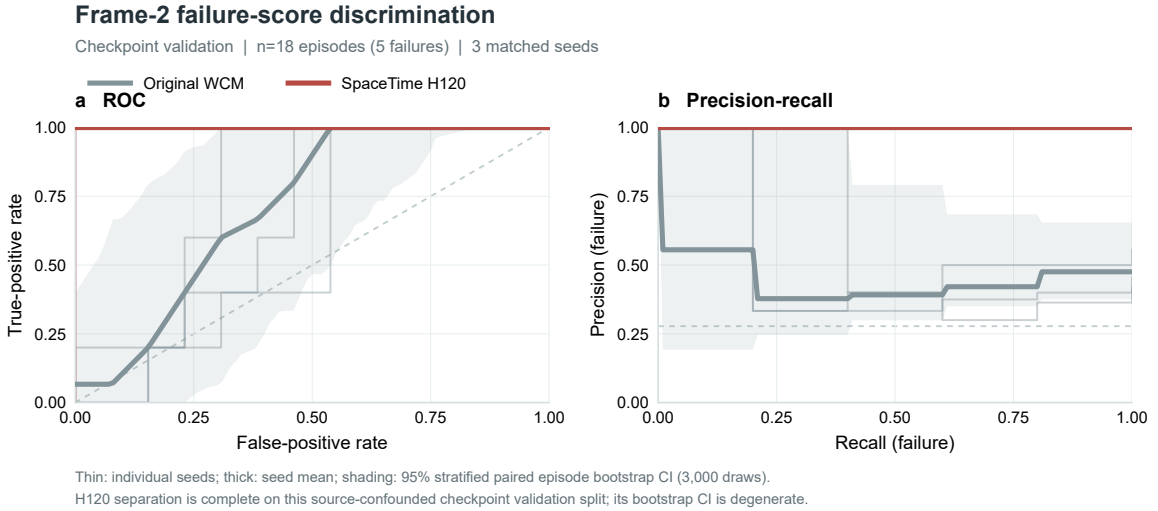


Figure 7: Threshold-free failure discrimination at frame 2. ROC and precision-recall curves compare the original WCM value score $-V(s)$ with the trained T8- H_{120} risk score on the same 18 validation episodes (five failures). Curves and shaded 95% bands reflect three model seeds and episode-level resampling; all resamples preserve the success/failure class counts. The same episodes recur across seeds.

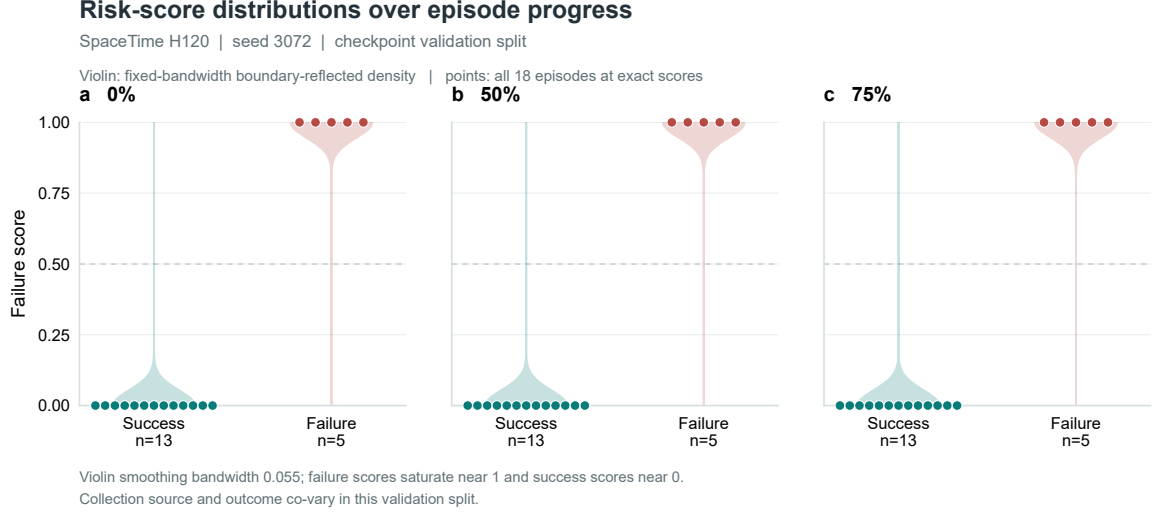


Figure 8: Failure-score distributions by true outcome. Violin summaries and individually jittered episode scores for true successes ($n = 13$) and true failures ($n = 5$) at the first, 50%, and 75% progress windows. All scores come from the T8- H_{120} risk head of seed 3072; the same 18 episodes appear in each panel. Progress fractions refer to each episode’s completed duration.

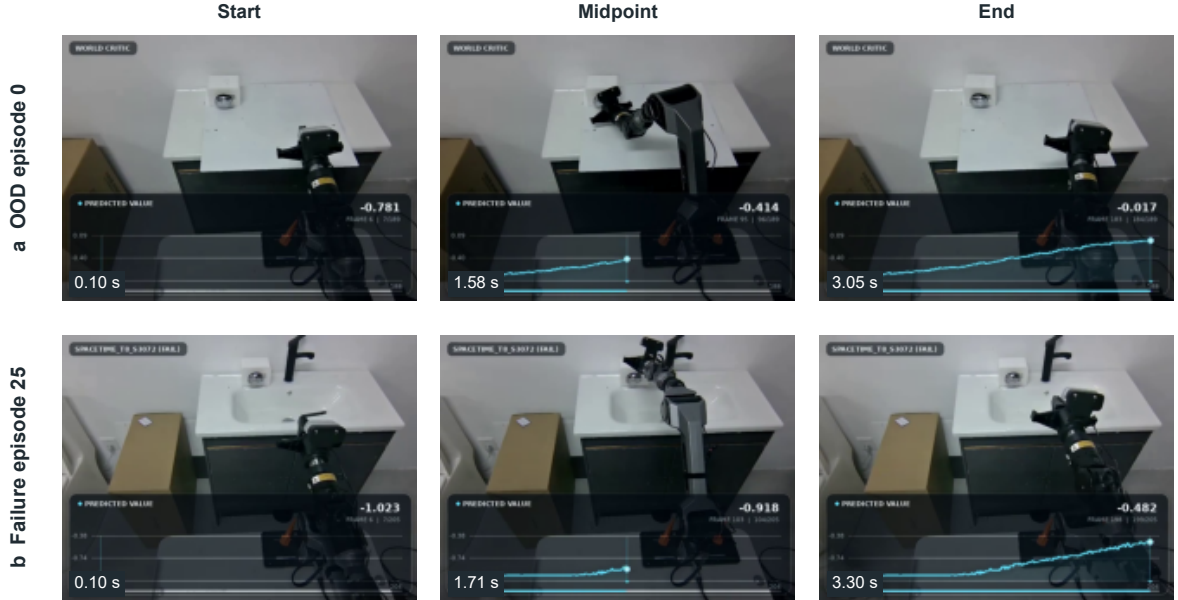


Figure 9: Illustrative value traces in rendered robot videos. Start, midpoint, and end frames from (a) an OOD episode rendered with T8- H_{120} and (b) a failure episode rendered with T8- H_{60} ; both use seed 3072. Video overlays display the prediction available up to each shown time. The failure example is from the expanded data collection and is not part of the all-success 0722 OOD set.

5 Discussion

The controlled span experiment shows that a longer sampled visual history can be used through the same 64-query interface in this robot value-learning setting. Its OOD MSE gain repeats across three seeds even though the number of frames, query tokens, and trainable blocks is unchanged. The original-WCM comparison adds evidence for the complete history-encoding system, while its different training recipe prevents a component-only interpretation.

The success/failure diagnostic shows strong separation on the checkpoint-validation cohort, including at an early fixed frame. Two features of this experiment constrain its interpretation. First, the original WCM score comes from a calibrated value output and the SPACETIME score comes from a supervised risk head. Second, outcome and collection source are associated in the 18-episode validation split. A source-matched, episode-disjoint mixed-outcome evaluation would distinguish task-outcome cues from collection appearance and test whether the observed separation transfers to new trajectories.

The fixed 64-query interface also has a computational cost: the T8 model has about 176.1 million parameters and an observed forward time of 12.9 ms per sample under the reported batch-four measurement, compared with 170.6 million parameters and 6.9 ms for original WCM. The current evidence covers one robot task family, two sampled spans, and an offline return target. It does not establish a monotone span scaling law or a guarantee that 64 queries retain all information in arbitrarily long histories.

6 Conclusion

SPACETIME compresses eight sampled two-camera observations into 64 learned queries for WCM value prediction. On the evaluated OOD episodes, increasing the history span from 60 to 120 sampled offsets lowered paired MSE in all three seeds without changing the frame or query count. The result identifies a useful operating range for fixed-query history encoding and motivates source-matched tests of outcome prediction and additional span settings.

A Input-mosaic control

An input-level mosaic was evaluated as a reference. The four-frame version places observations in a 2×2 canvas; the eight-frame version places matched H_{60} observations in a 2×4 canvas before a 224-pixel image processor. The eight-frame SPACETIME system lowered OOD MSE relative to the eight-frame mosaic in seeds 3072 and 42 by 0.01800 and 0.02455, respectively (paired 95% CIs $[-0.02065, -0.01538]$ and $[-0.02726, -0.02192]$). Pearson also increased by 0.02630 and 0.02942 with positive paired intervals. Mosaic frame count and canvas geometry are coupled in this comparison, so the result is a system reference rather than an isolated measurement of resolution loss.

References

- [1] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, et al. An image is worth 16x16 words: Transformers for image recognition at scale. In *International Conference on Learning Representations (ICLR)*, 2021.
- [2] Andrew Jaegle, Sebastian Borgeaud, Jean-Baptiste Alayrac, et al. Perceiver: General perception with iterative attention. In *International Conference on Machine Learning (ICML)*, 2021.
- [3] Alec Radford, Jong Wook Kim, Chris Hallacy, et al. Learning transferable visual models from natural language supervision. In *International Conference on Machine Learning (ICML)*, 2021.